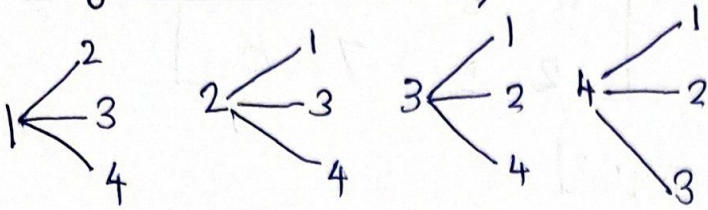


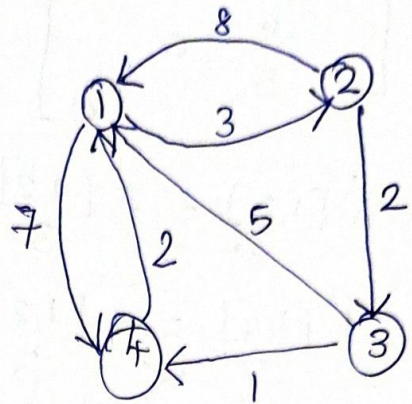
ALL PAIR SHORTEST PATH ALGORITHM

Using Dynamic Programming, FLOYD WARSHALL ALG.

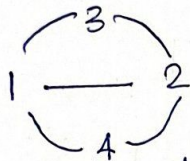


This is similar to single source shortest path (Dijkstra's Alg.)

$$n^2 \cdot n = O(n^3)$$



Solve using Dynamic Programming is done Sequence of decisions.



First Consider Vertex 1 as middle vertex.
All diagonals are zero because there is no self loop.
If there is no edge exist between any two vertices mark as ∞

$$A^0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 3 & \infty & 7 \\ 8 & 0 & 2 & \infty \\ 5 & \infty & 0 & 1 \\ 2 & \infty & \infty & 0 \end{bmatrix} \end{matrix}$$

$$A^1 = \begin{bmatrix} 0 & 3 & \infty & 7 \\ 8 & 0 & & \\ 5 & & 0 & 0 \\ 2 & & & \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 3 & \infty & 7 \\ 8 & 0 & 2 & 15 \\ 5 & 8 & 0 & 1 \\ 2 & 5 & \infty & 0 \end{bmatrix}$$

$$A^0[2,3] = A^0[2,1] + A^0[1,3]$$

$$2 < 8 + \infty$$

$$A^0[2,4] = A^0[2,1] + A^0[1,4]$$

$$\infty > 8 + 7$$

$$A^0[3,2] = A^0[3,1] + A^0[1,2]$$

$$\infty > 5 + 3 = 8$$

$$A^0[3,4] = A^0[3,1] + A^0[1,4]$$

$$\infty < 5 + 7$$

$$A^0[4,2] = A^0[4,1] + A^0[1,2]$$

$$\infty > 2 + 3 = 5$$

$$A^0[4,3] = A^0[4,1] + A^0[1,3]$$

$$\infty < 2 + \infty$$

$$A^2 = \begin{bmatrix} 0 & 3 & 2 & 15 \\ 8 & 0 & 8 & 0 \\ 5 & 8 & 0 & 0 \\ 2 & 5 & 7 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 3 & 5 & 7 \\ 8 & 0 & 2 & 15 \\ 5 & 8 & 0 & 1 \\ 2 & 5 & 7 & 0 \end{bmatrix}$$

$$A^1[1,3] = A^1[1,2] + A^1[2,3]$$

$$3 > 3 + 2 = 5$$

$$A^1[1,4] = A^1[1,2] + A^1[2,4]$$

$$7 < 3 + 15$$

$$A^1[3,1] = A^1[3,2] + A^1[2,1]$$

$$5 < 8 + 8 = 16$$

$$A^1[3,4] = A^1[3,2] + A^1[2,4]$$

$$1 < 8 + 15 = 23$$

$$A^1[4,1] = A^1[4,2] + A^1[2,1]$$

$$2 < 5 + 8$$

$$A^1[4,3] = A^1[4,2] + A^1[2,3]$$

$$5 > 5 + 2 = 7$$

$$A^3 = \begin{bmatrix} 0 & 5 & 7 \\ 0 & 2 & 1 \\ 5 & 8 & 0 \\ 7 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 3 & 5 & 6 \\ 7 & 0 & 2 & 3 \\ 5 & 8 & 0 & 1 \\ 2 & 5 & 7 & 0 \end{bmatrix}$$

$$A^2[1,2] < A^2[1,3] + A^2[3,2]$$

$$3 < 5 + 8$$

$$A^2[1,4] < A^2[1,3] + A^2[3,4]$$

$$7 < 5 + 1 = 6$$

$$A^2[2,1] > A^2[2,3] + A^2[3,1]$$

$$8 > 2 + 5 = 7$$

$$A^2[2,4] > A^2[2,3] + A^2[3,4]$$

$$15 > 2 + 1 = 3$$

$$A^2[4,1] > A^2[4,3] + A^2[3,1]$$

$$2 < 7 + 5$$

$$A^2[4,2] < A^2[4,3] + A^2[3,2]$$

$$5 < 7 + 8$$

$$A^4 = \begin{bmatrix} 0 & & & 6 \\ & 0 & & 3 \\ & & 0 & 1 \\ 2 & 5 & 7 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 3 & 5 & 6 \\ 5 & 0 & 2 & 3 \\ 3 & 6 & 0 & 1 \\ 2 & 5 & 7 & 0 \end{bmatrix}$$

$$\begin{aligned} A^3[1,2] &= 3 < A^3[1,4] + A^3[4,2] = 6 + 5 \\ A^3[1,3] &= 5 < A^3[1,4] + A^3[4,3] = 6 + 7 \\ A^3[2,1] &= 7 > A^3[2,4] + A^3[4,1] = 3 + 2 = 5 \\ A^3[2,3] &= 2 < A^3[2,4] + A^3[4,3] = 3 + 7 \\ A^3[3,1] &= 5 > A^3[3,4] + A^3[4,1] = 1 + 2 = 3 \\ A^3[3,2] &= 8 > A^3[3,4] + A^3[4,2] = 1 + 5 = 6 \end{aligned}$$

$$A^k[i,j] = \min \left\{ A^k[i,j], A^{k-1}[i,k] + A^{k-1}[k,j] \right\}$$

Deduce the Formula.

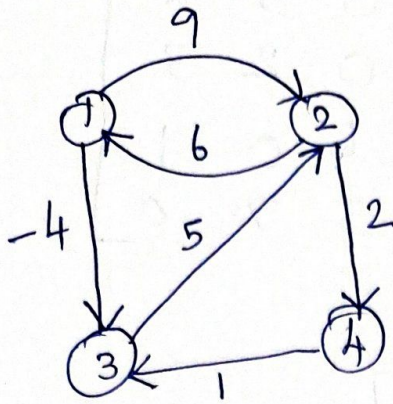
Code:

Generating Matrix →

```
for (k=1; k<=n; k++)
{
    for (i=1; i<=n; i++)
    {
        for (j=1; j<=n; j++)
        {
            A[i,j] = min(A[i,j], A[i,k] + A[k,j])
        }
    }
}
```

$O(n^3)$

Ex:



Soln.

$$A^0 = \begin{bmatrix} 0 & 9 & -4 & \infty \\ 6 & 0 & \infty & 2 \\ \infty & 5 & 0 & \infty \\ \infty & \infty & 1 & 0 \end{bmatrix}$$

$$A^1 = \begin{bmatrix} 0 & 9 & -4 & \infty \\ 6 & 0 & 2 & 2 \\ \infty & 5 & 0 & \infty \\ \infty & \infty & 1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 9 & -4 & 11 \\ 6 & 0 & 2 & 2 \\ 11 & 5 & 0 & 7 \\ \infty & \infty & 1 & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 0 & 1 & -4 & 3 \\ 6 & 0 & 2 & 2 \\ 11 & 5 & 0 & 7 \\ 12 & 6 & 1 & 0 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 0 & 1 & -4 & 3 \\ 6 & 0 & 2 & 2 \\ 11 & 5 & 0 & 7 \\ 12 & 6 & 1 & 0 \end{bmatrix}$$